

ECC (Simplified Elliptic Curve Demonstration)

1. Aim

To implement a simplified verification of whether a given point lies on an elliptic curve of the form $y^2 = x^3 + ax + b \pmod{p}$, and to understand the basic mathematical structure used in Elliptic Curve Cryptography (ECC).

2. Algorithm

1. Start the program.
2. Read the coordinates x and y of the point to be tested, entered by the user.
3. Read the curve parameters a and b entered by the user.
4. Read the prime modulus p entered by the user.
5. Compute the left-hand side value as $lhs = (y * y) \pmod{p}$.
6. Compute the right-hand side value as $rhs = (x^3 + a*x + b) \pmod{p}$.
7. Compare the values of lhs and rhs .
8. If lhs is equal to rhs , display that the point lies on the elliptic curve; otherwise, display that it does not, and stop the program.

3. Program

```
#include <stdio.h>

int main() {
    int x, y, a, b, p, lhs, rhs;

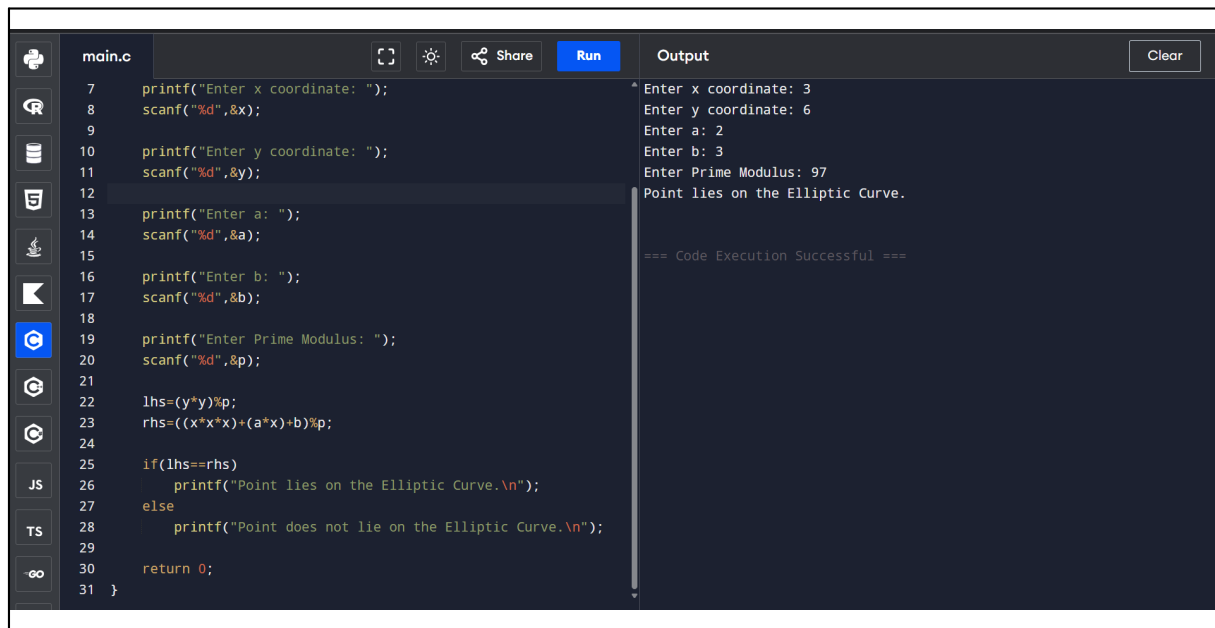
    printf("Enter x coordinate: ");
    scanf("%d", &x);
    printf("Enter y coordinate: ");
    scanf("%d", &y);
    printf("Enter a: ");
    scanf("%d", &a);
    printf("Enter b: ");
    scanf("%d", &b);
    printf("Enter Prime Modulus: ");
    scanf("%d", &p);

    lhs = (y * y) % p;
    rhs = ((x * x * x) + (a * x) + b) % p;

    if (lhs == rhs)
        printf("Point lies on the Elliptic Curve.\n");
    else
        printf("Point does not lie on the Elliptic Curve.\n");

    return 0;
}
```

4. Output Screenshot



The screenshot shows a code editor with a file named `main.c`. The code is in C and implements a program to verify if a point lies on an elliptic curve. The code prompts the user for the x and y coordinates of a point, the coefficients a and b of the curve, and a prime modulus p. It then calculates the left-hand side (lhs) and right-hand side (rhs) of the curve equation $y^2 \equiv x^3 + ax + b \pmod{p}$ and checks if they are equal. If they are, it prints "Point lies on the Elliptic Curve.", otherwise it prints "Point does not lie on the Elliptic Curve.". The output window on the right shows the program's execution with the following input: x=3, y=6, a=2, b=3, p=97. The output is "Point lies on the Elliptic Curve." and "=== Code Execution Successful ===".

```
7  printf("Enter x coordinate: ");
8  scanf("%d",&x);
9
10 printf("Enter y coordinate: ");
11 scanf("%d",&y);
12
13 printf("Enter a: ");
14 scanf("%d",&a);
15
16 printf("Enter b: ");
17 scanf("%d",&b);
18
19 printf("Enter Prime Modulus: ");
20 scanf("%d",&p);
21
22 lhs=(y*y)%p;
23 rhs=((x*x*x)+(a*x)+b)%p;
24
25 if(lhs==rhs)
26     printf("Point lies on the Elliptic Curve.\n");
27 else
28     printf("Point does not lie on the Elliptic Curve.\n");
29
30 return 0;
31 }
```

Output:

```
Enter x coordinate: 3
Enter y coordinate: 6
Enter a: 2
Enter b: 3
Enter Prime Modulus: 97
Point lies on the Elliptic Curve.

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter x coordinate: 3
Enter y coordinate: 6
Enter a: 2
Enter b: 3
Enter Prime Modulus: 17
Point lies on the Elliptic Curve.
```

6. Result

The program to verify whether a point lies on a given elliptic curve was implemented and executed successfully, confirming the point satisfied the curve equation under the given prime modulus.